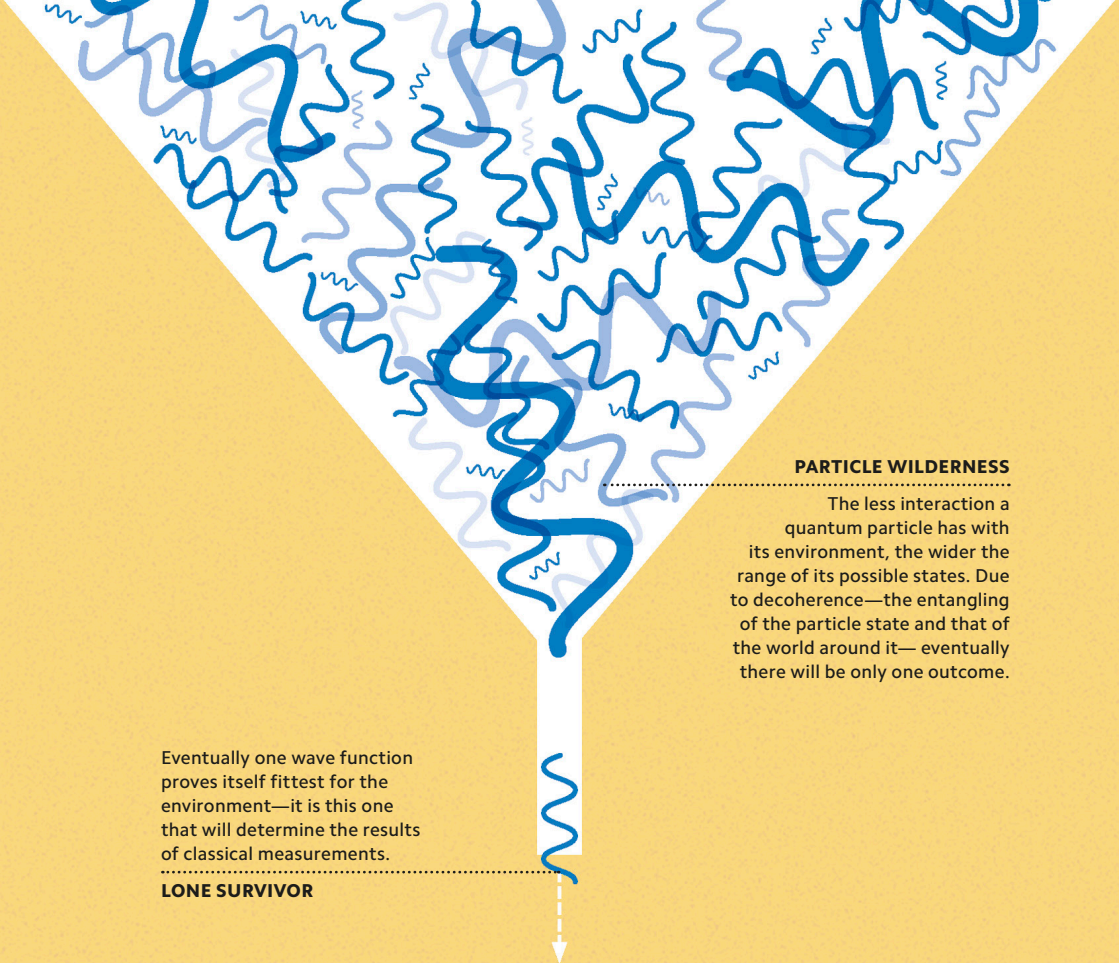




PARTICLE WILDERNESS

The less interaction a quantum particle has with its environment, the wider the range of its possible states. Due to decoherence—the entangling of the particle state and that of the world around it—eventually there will be only one outcome.

Eventually one wave function proves itself fittest for the environment—it is this one that will determine the results of classical measurements.

LONE SURVIVOR

SURVIVAL OF THE FITTEST

If hidden variables (see pp.56–57) do not guide the wave function to collapse into a certain state, perhaps something else does? Quantum Darwinism, as its name suggests, involves an idea similar to Charles Darwin’s evolutionary “survival of the fittest.” According to this interpretation, interactions between a particle and factors in its environment gradually filter the possible end states of its wave function until it settles on a single outcome known as a “pointer state.”